Aror University of Art, Architecture, Design & Heritage Sukkur

Linear Algebra Course Outline

BS Artificial Intelligence, Cyber Security and Multimedia Gaming Fall 2026

Course Title:	Linear Algebra
Course Code:	MAT-212
Credit Hours:	(3+0)
Course Instructors:	Mr. Rabnawaz Mallah, Mr. Sarfaraz Ahmed
E-mail:	Rabnawaz.faculty@aror.edu.pk, Sarfaraz.faculty@aror.edu.pk
Office location:	Faculty office, 2nd floor, Building 01
Consulting hours:	2:30PM to 4:30PM (Friday)

Course Description:

This course introduces the fundamental concepts of Linear Algebra, focusing on practical applications and problem-solving techniques. It is designed for students from non-mathematical backgrounds to develop a foundational understanding and apply these concepts in various fields.

COURSE LEARNING OUTCOMES

The students will be able:

CLO1: Understand and apply the basic concepts of Matrix, Determinants, and vectors.

CLO2: Utilize the concepts of matrices in daily livies..

CLO3: Analyze and interpret the properties of vectors with Eigen vectors.

Assessment

S. No	Assessment Activities	Percentage	Total Activities
1.	Sessional: Quizzes, Assignments, and Presentation	30%	2, 2,1
2.	Mid Term Exam	30%	1
3.	Final Exam	40%	1

Recommended Books:

1.	Elementary Linear Algebra	By Ron Larson	8th Edition
2.	Linear Algebra and its Applications	By Gilbert Strang	4th Edition
3.	Linear Algebra	By Lawrence E. Spence, Arnold Insel, Stephen H. Friedberg	5th Edition

Course Outline

Week No.	Lecture No.	TOPICS	REFERENCE	Course Covered
Week No: 01	LEC: 01-03	Introduction to System of Linear Equations Gaussian Elimination and Gauss-Jordan Elimination	Elementary Linear Algebra by Ron Larson	5%
Week No: 02	LEC: 04-06	Applications of Systems of Linear Equations Operations with Matrices	Elementary Linear Algebra by Ron Larson	10%
Week No: 03	LEC: 07-9	Properties of Matrix Operations The Inverse of a Matrix	Elementary Linear Algebra by Ron Larson	15%
Week No: 04	LEC: 10-12	Elementary Matrices The Determinant of a Matrix	Elementary Linear Algebra by Ron Larson	20%
Week No: 05	LEC: 12-15	Determinants and Elementary Operations Properties of Determinants	Elementary Linear Algebra by Ron Larson	30%
Week No: 06	LEC: 15-18	Introduction of Vectors Types of Vectors	Elementary Linear Algebra by Ron Larson	35%
Week No: 07	LEC: 18-21	Vectors Interpretation Vector Spaces	Elementary Linear Algebra by Ron Larson	40%

Week No: 08	LEC: 21-24	Subspaces of Vector Spaces Spanning Sets and Linear Independence	Elementary Linear Algebra by Ron Larson	50%
Mid Term				
Week No:09	LEC: 21-24	Basis and Dimension Rank of a Matrix and Systems of Linear Equations	Elementary Linear Algebra by Ron Larson	55%
Week No: 10	LEC: 25-27	Coordinates and Change of Basis Length and Dot Product in $\mathbb{R}^n$	Elementary Linear Algebra by Ron Larson	65%
Week No: 11	LEC: 28-30	Inner Product Spaces Orthonormal Bases: Gram-Schmidt Process	Elementary Linear Algebra by Ron Larson	75%
Week No:12	LEC: 31-33	Introduction to Linear Transformations The Kernel and Range of Linear Transformation	Elementary Linear Algebra by Ron Larson	80%
Week No: 13	LEC: 34-36	Matrices for Linear Transformations Transition Matrices and Similarity	Elementary Linear Algebra by Ron Larson	85%

Week No: 14	LEC: 37-39	Eigenvalues and Eigenvectors Diagonalization	Elementary Linear Algebra by Ron Larson	90%

Week No: 15	LEC: 39-42	Symmetric Matrices and Orthogonal Diagonalization	Elementary Linear Algebra by Ron Larson	95%
Week No: 16	LEC: 43-45	Applications of Eigenvalues and Eigenvectors	Elementary Linear Algebra by Ron Larson	100%